

Lecture # 2

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

TRUTH TABLES

$\sim P \wedge q$

P	q	$\sim P$	$\sim P \wedge q$
T	T	F	F
T	F	F	F
F	T	T	T
F	F	T	F

$\sim P \wedge (q \vee \sim r)$

P	q	r	$\sim P$	$\sim r$	$q \vee \sim r$	$\sim P \wedge (q \vee \sim r)$
T	T	T	F	F	T	F
T	T	F	F	T	T	F
T	F	T	F	F	F	F
T	F	F	F	T	T	F
F	T	T	T	F	T	T
F	T	F	T	T	T	T
F	F	T	T	F	F	F
F	F	F	T	T	T	T

$(P \vee q) \wedge \sim (P \wedge q)$

$$(P \vee Q) \wedge \sim(P \wedge Q)$$

P	Q	$P \vee Q$	$P \wedge Q$	$\sim(P \wedge Q)$	$(P \vee Q) \wedge \sim(P \wedge Q)$
T	T	T	T	F	F
T	F	T	F	T	T
F	T	T	F	T	T
F	F	F	F	T	F

Usage of 'OR':

Used in an exclusive sense (p or q, but not both)

As, Tomorrow at 9, I'll be Lahore or Islamabad.
↓

Here OR is used in exclusive sense bcz if both statements R true then we have F for the statement.

⇒ While defining a disjunction the word OR is used in its inclusive sense therefore the symbol ' $\vee$ ' means inclusive OR.

Exclusive OR:

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

Logical Equivalence:

If two logical expressions have the same logical values in the truth tables, then we say that the two

logical expressions are logically equivalent.

As, $\sim(\sim P)$ is logically equivalent P . or
 $\sim(\sim P) \equiv P$.

2ble -ve property: $\sim(\sim P) \equiv P$

Let $P =$ "I'm happy" then

$\sim P =$ "I am not happy".

and $\sim(\sim P) =$

"It is not true that I'm
 not happy".

Since $\sim(\sim P) \equiv P$ means given statement is
 equivalent to "I'm happy".

Ex: Show that $\sim(P \wedge q)$ and $\sim P \wedge \sim q$ are not
 logically equivalent.

P	q	$\sim P$	$\sim q$	$(P \wedge q)$	$\sim(P \wedge q)$	$\sim P \wedge \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	F	T	T
F	F	T	T	F	T	T

hence proved bcz truth values R different.

De Morgan's Law:

→ Negation of an AND statement is logically equivalent to the OR statement in which each component is negated.

Symbolically,

$$\sim(P \wedge q) \equiv \sim P \vee \sim q$$

→ Negation of an OR statement is logically equivalent to the AND statement in which each component is negated.

Symbolically,

$$\sim(P \vee q) \equiv \sim P \wedge \sim q$$

Truth table of $\sim(P \vee q) \equiv \sim P \wedge \sim q$

P	q	$\sim P$	$\sim q$	$P \vee q$	$\sim(P \vee q)$	$\sim P \wedge \sim q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

↓

Same truth values.

Application:

Give negation of the following statements:

(1) The fan is slow or it is very hot.

The fan is not slow and it's not very hot.

(2) Akram is unfit and Saleem is injured.

Akram is not unfit or Saleem is not injured.

Inequalities and De Morgan's law:

$$-1 < x \leq 4$$

↓ ↓ Use De Morgan's law to negate this inequality,
means $x > -1$ and $x \leq 4$

So its negation will be ($\because x > -1$ and $x \leq 4$)

$$x \leq -1 \text{ OR } x > 4.$$

We can also explain it as,

$$P: x > -1, \quad q: x \leq 4$$

$$\sim P: x \leq -1, \quad \sim q: x > 4$$

Negation of $x > -1$ and $x \leq 4 \equiv \sim(P \wedge q)$

by De Morgan's law,

$$\equiv \sim P \vee \sim q$$

$$\equiv x \leq -1 \text{ OR } x > 4$$

Ex: Show that $(P \wedge q) \wedge x \equiv P \wedge (q \wedge x)$ ($\because$ Associative law)

P	q	x	$(P \wedge q)$	$(P \wedge q) \wedge x$	$(q \wedge x)$	$P \wedge (q \wedge x)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	F	T	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

Are the statements $(P \wedge Q) \vee x$ and $P \wedge (Q \vee x)$ logically

P	Q	x	$P \wedge Q$	$(P \wedge Q) \vee x$	$(Q \vee x) \wedge P$	$P \wedge (Q \vee x)$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	F	T	T	T
T	F	F	F	F	F	F
F	T	T	F	T	T	F
F	T	F	F	F	T	F
F	F	T	F	T	T	F
F	F	F	F	F	F	F

So statements aren't logically equivalent.

Tautology: Statement form that is always true regardless of the truth values of statement variables.

Represented by "t".

Ex: Statement form $P \vee \sim P$ is Tautology.

P	$\sim P$	$P \vee \sim P$
T	F	T
F	T	T

$P \vee \sim P \equiv t$

Contradiction: Statement form that is always false regardless of truth values of statement variables.

Represented by 'c'

Ex: Statement form $P \wedge \sim P$ is contradiction.

P	$\sim P$	$P \wedge \sim P$
T	F	F
F	T	F

$$P \wedge \sim P \equiv C$$

REMARKS:

→ Most statements are neither tautology nor contradiction. → Negation of tautology is contradiction and vice versa. → Common usage we sometimes say that two statements are contradictory. [$\because$ means their conjunction is contradictory. (they can't both be true)]

Logical Equivalence Involving Tautology:

Ex: Show that $P \wedge t \equiv P$.

P	t	$P \wedge t$
T	T	T
F	T	F

$$P \wedge t \equiv P.$$

proved.

Involving Contradiction:

Show that $P \wedge C \equiv C$.

P	C	$P \wedge C$
T	F	F
F	F	F

$$P \wedge C \equiv C.$$

Exercise:

Show that (by using truth table)
 $(P \wedge Q) \vee (\sim P \vee (P \wedge \sim Q))$ is tautology

P	q	$\sim P$	$\sim q$	$P \wedge q$	$P \wedge \sim q$	$\sim P \vee (P \wedge \sim q)$	$(P \wedge q) \vee (\sim P \vee (P \wedge \sim q))$
T	T	F	F	T	F	F	T
T	F	F	T	F	T	T	T
F	T	T	F	F	F	T	T
F	F	T	T	F	F	T	T

Hence $(P \wedge q) \vee (\sim P \vee (P \wedge \sim q)) \equiv t$.

Exercise:

Use truth table to show that $(P \wedge q) \wedge (\sim P \vee q)$ is a contradiction.

P	q	$\sim P$	$\sim q$	$P \wedge \sim q$	$\sim P \vee q$	$(P \wedge \sim q) \wedge (\sim P \vee q)$
T	T	F	F	F	T	F
T	F	F	T	T	F	F
F	T	T	F	F	T	F
F	F	T	T	F	T	F

Hence proved,

$$(P \wedge \sim q) \wedge (\sim P \vee q) \equiv c.$$

Laws of Logic:

★ Commutative laws:

$$P \wedge q \equiv q \wedge P$$

$$P \vee q \equiv q \vee P$$

* Associative laws:

$$(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R)$$

$$(P \vee Q) \vee R \equiv P \vee (Q \vee R)$$

* Distributive laws:

$$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$$

$$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$$

* Identity laws:

$$P \wedge t \equiv P \quad \text{and} \quad P \vee c \equiv P$$

* Negation laws:

$$P \vee \sim P \equiv t$$

$$P \wedge \sim P \equiv c$$

* Double negation laws:

$$\sim(\sim P) \equiv P$$

* Idempotent laws:

$$P \wedge P \equiv P$$

$$P \vee P \equiv P$$

* De Morgan's laws: $\sim(P \wedge Q) \equiv \sim P \vee \sim Q$

$$\sim(P \vee Q) \equiv \sim P \wedge \sim Q$$

* Universal Bound laws:

$$P \vee t \equiv t$$

$$P \wedge c \equiv c$$

* Absorption laws:

$$P \vee (P \wedge Q) \equiv P$$

$$P \wedge (P \vee Q) \equiv P$$

★ Negation of t & c:

$$\sim t \equiv c$$

$$\sim c \equiv t$$

Lecture # 03

Laws of Logic

Using law of logic, simplify the statement form.

$$P \vee [\sim(\sim P \wedge q)]$$

$$P \vee [\sim(\sim P) \vee \sim q] \quad \text{De Morgan's law.}$$

$$P \vee [P \vee \sim q]$$

$$[P \vee P] \vee [\sim q] \quad \text{2ble negation law } \sim(\sim P) \equiv P$$

$$[P \vee P] \vee [\sim q]$$

$$P \vee \sim q \quad \text{Associative law for } \vee.$$

$$P \vee \sim q \quad \text{Idempotent law. } P \vee P \equiv P.$$

Using law of logic, verify the logical equivalence?

$$\sim(\sim P \wedge q) \wedge (P \vee q) \equiv P.$$

$$[\sim(\sim P) \vee (\sim q)] \wedge (P \vee q) \quad \text{De Morgan's law.}$$

$$(P \vee \sim q) \wedge (P \vee q) \quad \text{2ble negation law.}$$

$$P \vee (\sim q \wedge q) \quad \text{Distributive law.}$$

$$P \vee c$$

$$P \quad \text{Negation law.}$$

$$P \quad \text{Identity law.}$$

q	$\sim q$	$q \wedge \sim q$
T	F	F
F	T	F

Simplifying a Statement:

* You will get an A if u are hardworking and

the Sun shines, or you are hardworking and it rains."

let $p =$ "You are hardworking."

$q =$ "The sunshines."

$r =$ "It rains."

Condition is, $(p \wedge q) \vee (p \wedge r)$

Using distributive law in reverse,
 $= p \wedge (q \vee r)$ now putting back
into English, we can rephrase,

$\Rightarrow$ You will get an A if you are hardworking and
the sun shines or it rains.

Exercise:

1) It is not true that I'm tired and you are smart.
 $\sim (p \wedge q) = \sim p \vee \sim q$

I'm not tired or you are not smart.

2) It is not true that I'm tired or you are smart.
 $\sim (p \vee q) \Rightarrow \sim p \wedge \sim q$

I'm not tired and you are not smart.

3) I forgot my pen or my bag and I forgot my pen or my glasses.
 $(p \vee q) \wedge (p \vee r) \Rightarrow p \vee (q \wedge r)$

I forgot my pen or I forget my bag and glasses.

(4) It is raining and I have forgotten my umbrella or it is raining and I have forgotten my hat.

$$P \wedge (Q \vee R)$$

It is raining and I have forgotten my umbrella or my hat.

Conditional Statements:

"If you earn A in Math, then I'll buy you a computer."

↓

Making of 2 simple statements,

P = "You earn A in Math"

Q = "I'll buy you a computer"

→ Original statement then saying:

If P is true then Q is true or more simply, if P, then Q, also rephrase this P implies Q.

→ Represented by $P \rightarrow Q$.

Conditional Statements OR Implication:

If p and q are statements Variables, the conditional of q by p is "If p then q" or "p implies q"

and is denoted by $P \rightarrow Q$.

it's false when P is true and Q is false.
" $\rightarrow$ " is called conditional operator.

$P \rightarrow Q$, P statement is called HYPOTHESIS or ANTECEDENT.

and Q is called CONCLUSION or CONSEQUENT.
Truth Table,

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Practice, Determine the truth values of following conditional statements.

- * "If $1 = 1$ then $3 = 3$." $\rightarrow$ TRUE
- * "If $1 = 1$ then $2 = 3$." $\rightarrow$ FALSE
- * "If $1 = 0$ then $3 = 3$." $\rightarrow$ TRUE
- * "If $1 = 2$ then $2 = 3$." $\rightarrow$ TRUE
- * "If $1 = 1$ then $1 = 2$ and $2 = 3$." $\rightarrow$ FALSE
- * "If $1 = 3$ or $1 = 2$ then $3 = 3$." $\rightarrow$ TRUE

Alternating ways of expressing Implication,

Implication $p \rightarrow q$ could be expressed in many alternative ways are:

"If p then q "

" p implies q "

"not p unless q "

" q follows from p "

"If p , q "

" q if p "

" p only if q "

" q wherever p "

" p is sufficient for q "

" q is necessary for p "

Exercise.

→ Your guarantee is good only if you bought your CD less than 90 days ago.

If your guarantee is good then you must have bought your CD less than 90 days ago.

→ To get tenure as a professor, it's sufficient to be world-famous.

If you are world-famous, then you will get tenure as a professor.

→ That you get the job implies that you have the best credentials.

If you get the job, then you have the best credential.

→ It is necessary to walk 8 miles to get to the top of the peak.

If you get to the top of the peak, then you must have walked 8 miles.

Translating English Sentences to Symbols:

Let p and q be the propositions:

p = "You get an A on the final exams."

q = "You do every exercise in this book."

r = "You get an A in this class."

→ To get an A in this class it's necessary for you to get an A on the final. $P \rightarrow r$

→ You do every exercise in this book; you get an A on the final exams, implies you get an A in this class.

$$P \wedge q \rightarrow r$$

Translating Symbolic proposition to English:

Let p , q and r be the propositions:

p = "You have the flu."

q = "You miss the final exam."

r = "You pass the course."

→ $P \rightarrow q$

If you have the flu, then you will miss the final exam.

→ $\neg q \rightarrow r$

If you don't miss the final exam, then you will pass the course.

→ $\neg p \wedge \neg q \rightarrow r$

If you neither have the flu nor miss the final exam, then you pass the course.

Hierarchy of operations for logical connectives:

→ Ex

Construct a table for statement form $(P \rightarrow q) \wedge (\sim p \rightarrow r)$

P	q	r	$P \rightarrow q$	$\sim P$	$\sim P \rightarrow r$	$(P \rightarrow q) \wedge (\sim P \rightarrow r)$
T	T	T	T	F	T	T
T	T	F	T	F	T	T
T	F	T	F	F	T	F
T	F	F	F	F	T	F
F	T	T	T	T	T	T
F	T	F	T	T	F	F
F	F	T	T	T	T	T
F	F	F	T	T	F	F

→ $P \vee \sim q \rightarrow \sim P$

P	q	$\sim P$	$\sim q$	$P \vee \sim q$	$P \vee \sim q \rightarrow \sim P$
T	T	F	F	T	F
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

→ $P \rightarrow q \equiv \sim q \rightarrow \sim P$

P	q	$\sim P$	$\sim q$	$P \rightarrow q$	$\sim q \rightarrow \sim P$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Same truth values hence given 2 expressions are logical equivalent.

Implication law.

$$P \rightarrow q \equiv \sim P \vee q$$

P	q	$P \rightarrow q$	$\sim P$	$\sim P \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Same truth values so it's a logical equivalent.

Negation of a conditional statement:

Since $P \rightarrow q \equiv \sim P \vee q$

So $\sim(P \rightarrow q) \equiv \sim(\sim P) \wedge (\sim q)$ by De Morgan's law,
 $\equiv P \wedge \sim q$ by double negation law.

Means, the negation of "if p then q" is logically equivalent to "p and not q".

Accordingly, the negation of "if-then" statement doesn't start with "If".

Examples:

→ If Ali lives in Pakistan then he lives in Lahore.

Ali lives in Pakistan and he doesn't live in Lahore.

→ If x is prime then x is odd or x is 2.

x is prime but x is not odd and x isn't 2.

→ If m is divisible by 6, then m is divisible by 2 and m is divisible by 3.

m is divisible by 6 but m isn't divisible by 2 or by 3.

→ If my car is in the repair, then I can't get to class.

My car is in the repair shop and I can get to class.

Inverse of a Conditional Statement:

→ Inverse of the conditional statement $P \rightarrow q$ is $\neg P \rightarrow \neg q$.

→ Conditional and its inverse aren't logical equivalents.

P	q	$P \rightarrow q$	$\neg P$	$\neg q$	$\neg P \rightarrow \neg q$
T	T	T	F	F	T
T	F	F	F	T	T
F	T	T	T	F	F
F	F	T	T	T	T

↓
Different truth values.

Writing Inverse.

→ If today is Friday, then $2+3=5$

If today isn't Friday, then $2+3 \neq 5$.

→ If P is a square, then P is a rectangle.

If P isn't square then P isn't a rectangle.

Converse of a Conditional Statement:

→ Converse of conditional statement $p \rightarrow q$ is $q \rightarrow p$.

→ Conditional and its converse aren't equivalent.

P	q	$P \rightarrow q$	$q \rightarrow P$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

↓ Not same.

Writing Converse:

→ If today is Friday, then $2+3=5$.

If $2+3=5$ then today is Friday.

→ If P is a square, then P is rectangle.

If P is a rectangle, then P is square.

Contrapositive of a Conditional Statement:

Contrapositive of a conditional statement is, $p \rightarrow q$

$$\neg q \rightarrow \neg p$$

Conditional and its contravene are equivalent.
Symbolically,

$$P \rightarrow Q \equiv \sim Q \rightarrow \sim P$$

→ If today is Friday, then $2+3=5$

If $2+3 \neq 5$, then today isn't Friday.

→ If my car is in the repair shop, then I can't get to class.

If I can get the class, then my car isn't in the repair shop.

Exercise,

Same values.

* $P \rightarrow Q \equiv \sim Q \rightarrow \sim P$

P	Q	$P \rightarrow Q$	$\sim P$	$\sim Q$	$\sim Q \rightarrow \sim P$
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T

* $Q \rightarrow P \equiv \sim P \rightarrow \sim Q$

Q	P	$Q \rightarrow P$	$\sim P$	$\sim Q$	$\sim P \rightarrow \sim Q$
T	T	T	F	F	T
F	T	T	F	T	T
T	F	F	T	F	F
F	F	T	T	T	T

Same values.